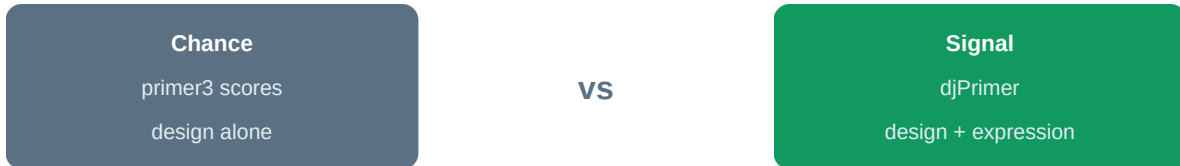


djPrimer

Design with primer3. Prioritize with this.

An expression-aware model that predicts which designed qPCR assays will actually validate — far better than any thermodynamic design score.



predicting validated outcomes · cross-validated by gene

Primer design software such as primer3 is excellent at what it does: it returns primer pairs that satisfy a set of thermodynamic constraints. But a well-formed primer pair and a working assay are not the same thing, and the gap between them is paid for in wet-lab cycles: reagents, instrument time, and weeks spent validating assays that were never going to fire.

We measured that gap on over **2,800** real primer/probe sets, each carried through validation across over **300** cell lines. primer3's own design scores predict whether an assay actually works no better than a coin flip. A model that adds one thing primer3 never sees — how much of the target is present — predicts success far more reliably, and as a triage tool it avoids roughly half of all failures while setting aside only about one good assay in ten. We call that model **djPrimer**.

primer3 is not the competitor here. It is the starting point. This model does not design better primers; it predicts, far better than any design score can, which designed assays will succeed.

01 · THE PROBLEM

A design score is not a success prediction

Every qPCR program runs the same loop: pick targets, design primers, order oligos, validate at the bench, keep what works. The validation step is where the cost lives, and a meaningful fraction of assays fail there. The natural hope is that the design score already tells you which ones.

It does not, once the primers are professionally designed, and the reason is almost obvious on reflection. A design tool's job is to reject bad primers, so by the time a pair reaches your order form it already clears every thermodynamic bar. The scores that separated good designs from bad ones during design are now uniform across everything you actually run, and a score that is the same for winners and losers cannot tell them apart.

02 · THE EVIDENCE

Measured on thousands of real assays

We used a validation database of over 2,800 primer/probe sets, each with a recorded validation outcome across over 300 cell lines: over 100,000 assay results in total. We reduced it to the assays whose outcome is unambiguous, reconstructed the actual amplicon each pair produces, and asked a series of prediction questions under honest cross-validation, grouped by gene so no gene is used to predict itself.

primer3's own scores predict success at chance. Scoring every primer with primer3's own thermodynamic functions and predicting success from them leaves the prediction at chance. Adding the reconstructed amplicon's composition and the local template structure adds nothing. In this set of professionally designed primers, the sequence simply does not carry the answer.

Target expression does. Bringing in how broadly and highly each target gene is expressed — from public RNA-seq references entirely independent of the qPCR data — predicts success well above chance on its own. Combined with the primer3 baseline, the full model does better still. Every bit of the improvement is expression; the sequence features contribute almost nothing.

Why it works, in one line: the assays that never validate overwhelmingly target genes that are not expressed in the panel. A low djPrimer score for a well-designed primer is usually the model telling you the target is silent in your intended sample.

03 · HEAD TO HEAD

As a decision tool, not just a statistic

Rank your candidate assays by a score and set aside the lowest-scoring quarter before committing them to the bench. What happens?

Prioritized by	Failures avoided	Good assays set aside	Success rate of what you run
djPrimer	about half	few	clearly higher
primer3 score	far fewer	many	essentially unchanged

The model roughly halves your failed validations while barely touching the assays that would have worked. primer3's score, applied the same way, removes almost as many good assays as bad ones and leaves your success rate essentially where it started, because that score is close to uninformative about the real outcome. Ranked by the model, the assays at the bottom of the list rarely work and the ones at the top usually do. That is a usable triage signal. A design score is not.

04 · IN PRACTICE

What this means for your lab

Triage before the bench. Run the high-scoring assays first, and deprioritize or redesign the bottom of the list — avoiding about half of failed validations for the cost of re-examining a small slice of good ones.

Catch silent-target failures early. A low score for a well-designed primer is almost always the model telling you the target is not expressed in your intended sample — a redesign or sample-choice decision made before you spend reagents, not after.

Rank a large panel. When you have more candidate assays than bench capacity, an expression-aware score puts the ones most likely to work at the top. You still design with primer3, then score the designs with this model and let the ranking decide where your effort goes.

```
# djPrimer: score any gene + primer pair for its success probability
bin/djprimer GAPDH CAACAGTGGCAACACCTTGTG TGGGTTGGTCATGCTCACTAG
```

05 · THE FINE PRINT

Scope and honest limitations

- **A prioritizer, not an oracle.** The score is a strong triage signal, not a guarantee. It reorders your queue; it does not certify individual assays.
- **The comparison is on outcome prediction, not design.** primer3 was never built to predict validation outcomes. We claim only that its scores — sometimes leaned on as a proxy for “this assay will work” — do not carry that information once the primers are already good.
- **The model predicts intrinsic success across a panel,** because its expression features are per-gene. Feeding it expression matched to your specific cell line or tissue would sharpen it further, and is the natural next step for a production deployment.
- **Specificity is the lever we did not exhaust.** Off-target priming needs sequence alignment rather than thermodynamics. It is the most likely source of the small residual signal we could not explain, and a good target for a future model version.

SUMMARY

Design with primer3. Prioritize with this.

The distance between a well-designed primer and a working assay is target biology, and primer3 cannot see it. Measured on thousands of real assays, primer3's design scores predict validation success at chance; an expression-aware model predicts it far more reliably and, used as a triage tool, roughly halves failed validations while barely touching the good assays. Spend your bench time on the assays that will work.